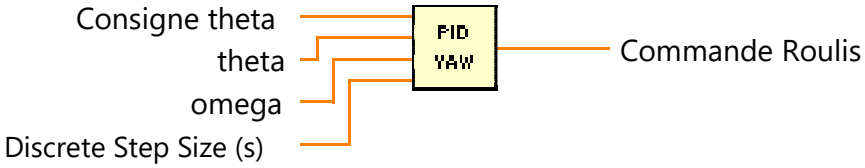
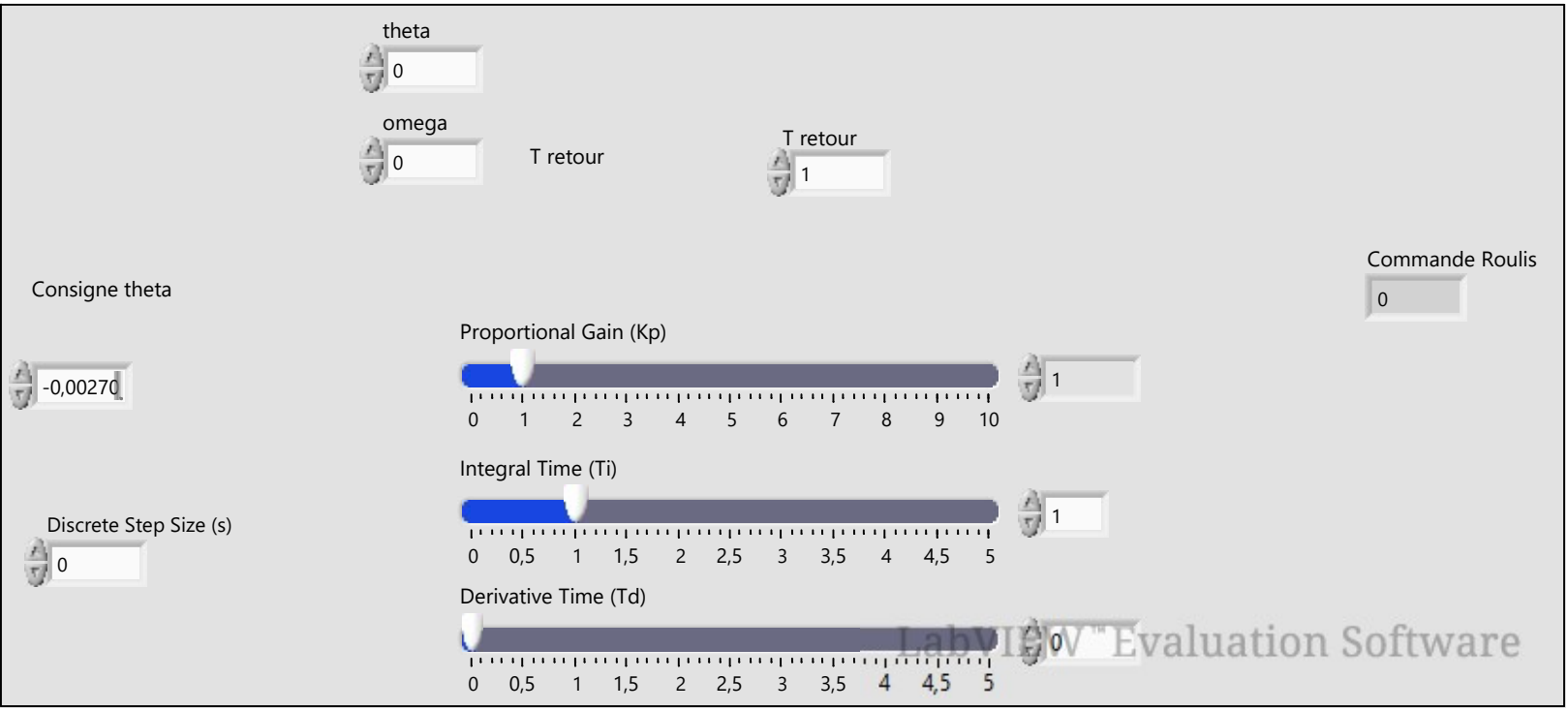
PID Yaw.vi





Consigne theta

theta

Discrete Step Size (s)

omega

Proportional Gain (Kp)
<B>Proportional Gain (Kc)</B> represents the proportional gain of the controller.

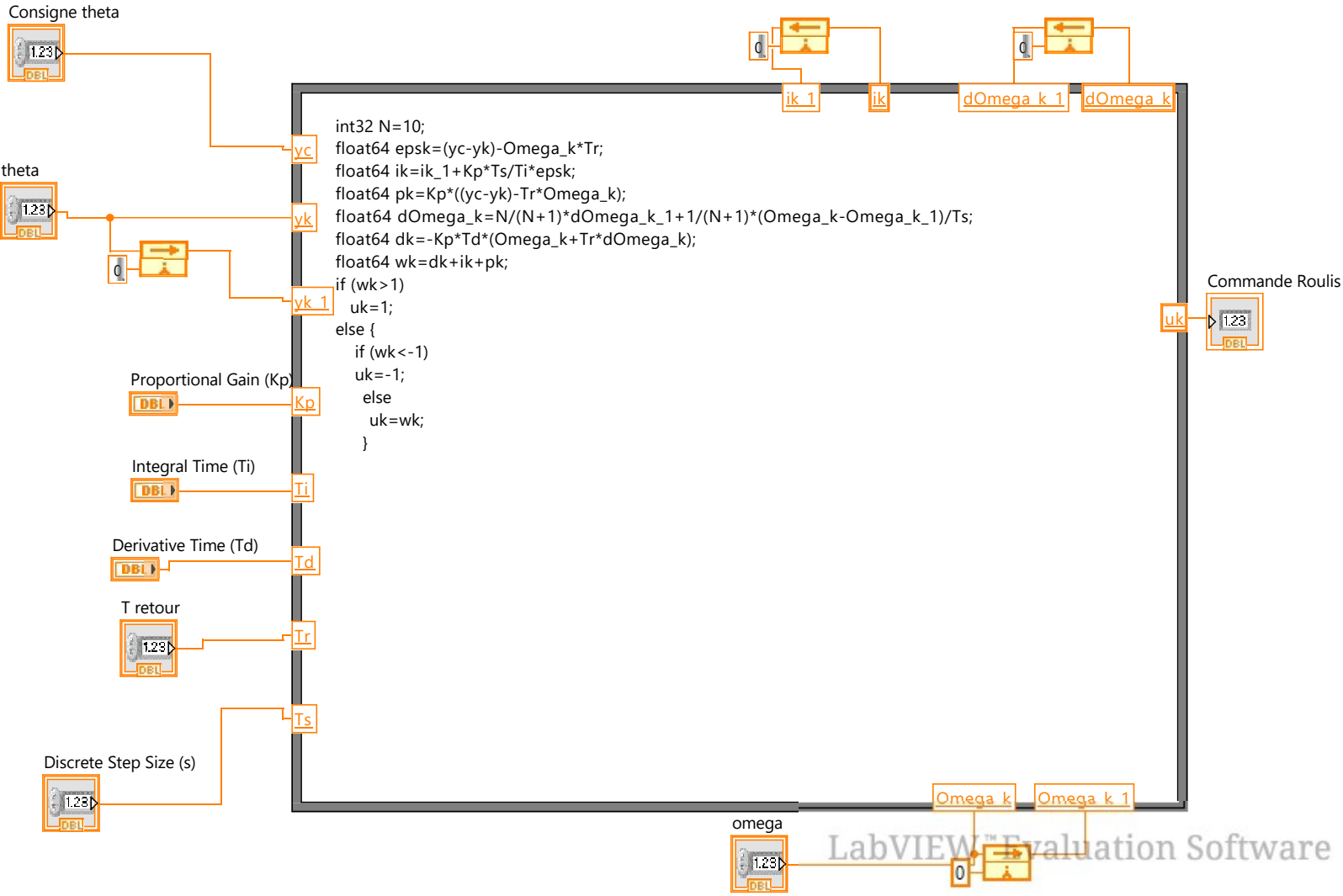
Integral Time (Ti)
<B>Integral Gain (Ki)</B> is the controller parameter that adjusts the effect of the error integral term E(s)/s on the controller output U(s).

Derivative Time (Td)
<B>Derivative Gain (Kd)</B> is the controller parameter that adjusts the effect of the error derivative term sE(s) on the controller output U(s).

T retour[T retour]
T retour

Commande Roulis

PID YAW


Consigne theta
theta
Proportional Gain (Kp)
Integral Time (Ti)
Derivative Time (Td)
T retour
Discrete Step Size (s)
yc
yk
yk_1
Kp
Ti
Td
Tr
Ts
ik_1
ik
dOmega_k_1
dOmega_k
int32 N=10;
float64 epsk=(yc-yk)-Omega_k*Tr;
float64 ik=ik_1+Kp*Ts/Ti*epsk;
float64 pk=Kp*((yc-yk)-Tr*Omega_k);
float64 dOmega_k=N/(N+1)*dOmega_k_1+1/(N+1)*(Omega_k-Omega_k_1)/Ts;
float64 dk=-Kp*Td*(Omega_k+Tr*dOmega_k);
float64 wk=dk+ik+pk;
if (wk>1)
uk=1;
else {
if (wk<-1)
uk=-1;
else
uk=wk;
}
uk
Commande Roulis
omega
Omega_k
Omega_k_1
LabVIEW™ Evaluation Software